CODE SPORT

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In this month's column, we discuss a medley of computer science interview questions.

Last month, we had discussed named entity recognition and relation extraction problems in natural language processing. In this month's column, we take a break from this topic and feature a bunch of interview questions on algorithms, data structures and machine learning.

1. You are given a data set for a binary classification problem. The input data is a set of features associated with that problem and the ground truth is a binary label 0/1. Your initial solution gave you a test set F1 score of 0.8. In order to improve this F1 score, you decided to try to improve the performance by creating additional features – using polynomial functions of the input variables. You expected your new solution to give a better F1 test set score. But you got only 0.69. Can you explain what may be causing this?
2. How do you decide what may be a good activation function for the deep learning model you are building? There are a number of activation functions – RelU, Tanh, Leaky RelU, Swish etc. How do you choose the right activation function?
3. What is a bagging classifier and what is a stacking classifier? If you have a handful of weak base classifiers for a binary classification problem, will you combine them using a bagging classifier or a stacking classifier? Explain the reason behind your choice.
4. Let us assume that you decided to use a bagging classifier for a binary classification problem. You decided to combine the classifiers using majority voting. Will you combine them using majority voting on the binary labels or by using majority voting on the class probabilities? Or will you use majority voting on the weighted average of class probabilities? Explain the reason behind your choice.
5. We all know about classifier ensembles. You are given only one neural network model based on, say, BERT. Now you are asked to train this model and build an ensemble classifier. Explain how you can do that. This type of ensemble classifier built using a single neural network topology is known as snapshot ensemble classifier.
6. Explain the K-fold cross validation and stratified cross validation approaches. When will you choose one over the other? And how does one choose the value of K in K-fold cross validation?
7. When designing a convolutional neural network (very often used for image processing problems) layer, in addition to filter sizes, there are two additional hyper parameters – namely, stride and padding. Can you explain what these parameters are and how to select their value?
8. We are all familiar with the stochastic gradient descent method for training neural networks. Let us assume that you have two training schedules, one where the batch size is 32 samples and another where it is 256 samples. Which of these training schedules will converge faster (i.e., will require lesser number of epochs to reach the specified loss threshold)? Can you explain the reason behind your answer?
9. Can you explain the advantage of a masked language model as opposed to a standard left to right language model?
10. You have been given all the stock market transactions made by a user in his stock exchange account, while buying and selling shares. Each transaction consists of the details on: (a) name of the company, (b) type of transaction — buy/sell, (c) number of shares bought/sold, (d) price of each share bought/sold, and (e) date of the transaction. You also have access to stock market information in terms of the price of each company's stock at the end of every day. The user would like to know which of his shares he can sell at a profit of 10 per cent or above on each day. You are asked to write a program, which examines the closing price